

Next, in the “Reduce” step, the workers process each group of data, per key, in parallel.

A set of ‘reducers’ can also perform the reduction in parallel, provided that all outputs of the map operation that share the same key are presented to the same reducer at the same time. While this process can often appear inefficient compared to algorithms that are traditionally more sequential (since the reduction process need not be broken down then), MapReduce can be applied to much larger datasets. The parallelism also allows a possibility of recovering from partial failure of jobs during the operation. i.e. if one mapper or reducer fails, the work can be reassigned to another one. In many situations, the input data might already be distributed among many different servers, in which case the map job becomes greatly simplified by assigning map servers that would process the locally present input data.

AWS provides a fully managed framework that makes it easy, fast, and cost-effective to process large amounts of data across scalable Amazon EC2 instances. You can also run popular distributed frameworks such as Apache Spark, HBase, Presto, and Flink in Amazon EMR, as well as interact with data in other AWS data stores such as Amazon S3 and Amazon DynamoDB.

Amazon EMR can be easily applied to a number of big data applications, including log analysis, web indexing, data transformations (ETL), machine learning, financial analysis, scientific simulation, and bio-informatics. An EMR cluster can be set up on AWS in a matter of minutes. Let us take a look at how we can do that.

Setting up an EMR Cluster on AWS:

- Navigate to ‘EMR’ from the ‘Services’ tab. On opening the EMR console, you will be presented with a home screen that lists your current clusters. If there aren’t any, click on the “Get Started Now” button.
- Click on “Create Cluster” to bring up the wizard that allows you to create



Create your EMR cluster.

an EMR cluster. You will be asked to provide a cluster name, as well as the choice of starting up a cluster or a complete step function that would start the cluster, run the job and terminate it on completion. Let’s go with the second option.